

Experiment 11: Implement a Simple Top-Down Parser for Context-Free Grammars

Program

```
grammar = {  
    "S": [["a", "A"]],  
    "A": [["b"]]  
}  
  
def parse(symbol, tokens):  
    if not tokens:  
        return False  
  
    if symbol not in grammar:  
        return tokens[0] == symbol and tokens[1:]  
  
    for production in grammar[symbol]:  
        remaining = tokens  
        success = True  
  
        for sym in production:  
            result = parse(sym, remaining)  
  
            if result is False:  
                success = False
```

```
break
```

```
remaining = result
```

```
if success:
```

```
    return remaining
```

```
return False
```

```
sentence = input("Enter sentence (space separated): ").split()
```

```
result = parse("S", sentence)
```

```
if result == []:
```

```
    print("Sentence Accepted")
```

```
else:
```

```
    print("Sentence Rejected")
```

Output

```
        remaining = result

    if success:
        return remaining

    return False

sentence = input("Enter sentence (space separated): ").split()

result = parse("S", sentence)

if result == []:
    print("Sentence Accepted")
else:
    print("Sentence Rejected")
```

Enter sentence (space separated): a b
Sentence Accepted

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